# ForgeGIS™

## *GPU-Accelerated Geospatial Primitives, Native to MCP*

An MCP server that gives an agent real geospatial compute: viewsheds, watersheds, slope, interpolation, overlay and topology, point clouds, routing — 237 of its 239 catalogued operations reachable, on the GPU where the algorithm allows and the CPU where it does not. The agent starts with a nine-tool surface and composes what it needs into one call. Failures come back as typed reasons it can branch on, and every answer is the one a Java script calling the same operation would get.

**9 tools**

**the default surface**

*enough to list 170 composable ops and run a chain or a branching graph of them in one call*

**84 tools**

**under the standard profile**

*5 primitives, 75 typed intent tools and 4 pipeline tools; a hidden tool stays callable by name*

**Typed**

**errors an agent can branch on**

*error.category and error.reason from a closed vocabulary, frozen in a versioned contract*

### Compose, or call

Two ways in, one engine underneath. `list_ops` browses 170 composable ops; `run_pipeline`, `run_vector_pipeline` and `run_dag` run a chain or a branching graph of them in one call, with raster intermediates staying on the GPU. Or call one of 75 typed intent tools — viewshed, hydrology, interpolation, point-cloud processing, geodesic distance — with a strict input schema.

### Small by default

The default `minimal` profile advertises nine tools, about 2.7k tokens of tool list, so an agent stays inside its context budget; `--profile standard` or `FORGEGIS_MCP_PROFILE` advertises 84, about 19k. The profile governs discovery, not capability: anything the agent already knows the name of is callable under either profile.

### The agent's answer is the script's

The MCP layer marshals parameters onto the same implementations the Java API calls, so a result depends on the operation and its inputs, not on the caller. Every measurement is on the WGS84 ellipsoid and names its model. Random methods default to a fixed seed, and tiled runs are byte-identical to untiled, so a rerun repeats.

**New in 1.2.6 — answers an agent can stand behind.** Every length, area, distance and bearing comes from one geodesic engine on the WGS84 ellipsoid (Karney's method), so "within 5 km" means 5 km in any coordinate system. Outputs with a GDAL counterpart are checked against it — contours within 1 cm of `gdal_contour` on real SRTM data, NoData filling on GDAL's inverse-distance method — and all 130 raster ops are held to CPU/GPU agreement and a NoData sweep, so an answer agrees, within each op's declared tolerance, whichever device ran it and however the voids were encoded.


CONTINUED ON REVERSE →

WHO THIS IS FOR

## *Agent platforms grounding spatial reasoning in real geometry*

Agents that must answer what is visible from a point, where water goes, how far apart two features really are, or what overlaps what — and need a computed answer, not a guess. ForgeGIS takes the coordinate systems, predicates and pipeline composition off the agent's hands: point parameters are read as longitude and latitude unless the call says otherwise.

## *MCP-native orchestration*

Any MCP client can spawn the server over stdio. In the Forge suite, ForgeMind™ runs it as a tool server beside ForgeData™, which finds the data; outside it, one shaded jar is the whole deployment. File access is confined to configured read and write roots, every call has a timeout, and results are bounded before they reach a language model.

**Errors worth branching on.** Every failure carries an `error.category` — `TOOL_ERROR`, `SANDBOX_DENIED`, `TOOL_TIMEOUT`, `SERVER_BUSY`, `CANCELLED` — and where it matters an `error.reason`: `INSUFFICIENT_HEAP`, `INSUFFICIENT_GPU_MEMORY`, `INPUT_TOO_LARGE`, `POINT_OUTSIDE`, `SOURCE_OUTSIDE`, `DESTINATION_OUTSIDE`, `SOURCE_NODATA`, and sixteen sandbox reasons. A point off the raster is refused, never quietly clamped.

**Refused before it runs (new in 1.2.6).** Every pipeline, graph and single-op call compares the operation's measured heap demand with a budget of half the free heap. If it will not fit, the call returns `INSUFFICIENT_HEAP` and the op never runs — nothing is allocated. An agent can shrink the area or split the job instead of waiting on a crash.

**A contract you can pin.** The whole MCP surface — every tool's schema and defaults, every op's parameters, the output keys and the error vocabulary — is recorded in a versioned contract snapshot (1.1.0), and the build fails on an unversioned change. `contract_info` tells a running client which version it is talking to.

**Changes in 1.2.6 an agent integration must act on.** `compute_haversine_distance` is now `compute_geodesic_distance`, with no alias — the old name is an unknown tool, and the default answer moves from the sphere to WGS84. Viewshed output is UINT8: 255 visible, 0 not visible, DEM voids tagged NoData. Every point parameter defaults to EPSG:4326.

**Fast enough to sit in a loop.** Measured on 1.2.6, October 2026, on one laptop: multi-stage pipelines ran 11–182× ahead of the same chain as GDAL command-line tools at 4096² rasters, and operations with a GDAL equivalent 15× on geometric mean (warm medians, 242 records from 16×16 to 4096×4096 and real elevation tiles, 239 faster, 3 slower). The benchmark's timing method was tightened in this release, so earlier figures are not directly comparable.

## Build with ForgeGIS

The Technical Brief covers the agent surface and profiles, the error vocabulary, the accuracy evidence, the capability catalog and deployment. ForgeGIS is proprietary and licensed by written agreement; evaluation builds and Maven coordinates are available to qualified teams.

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